Exploring Human-Centered Design Barriers in AI Powered MVPs: A Study from Emerging Market Startups

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Abstract*— This research examines the challenges designers face when adhering to human-centered design principles during the design and implementation of AI-based Minimum Viable Products (MVPs) in digital startups across Pakistan. Startups in emerging economies utilize tools such as chatbots, recommendation engines, and predictive software in the initial stages of product development. A lack of advancement in AI design, issues related to data quality, and infrastructure limitations can impede the delivery of functional and ethical AI based solution. A survey of 108 expert professionals identified integration issues between AI models and frontend design, limited availability of quality training data, low digital or AI literacy among target users, High cost of AI tools or platforms and balancing simplicity with AI functionality as the primary obstacles. The findings suggest that interdisciplinary collaboration, understanding of user needs, feedback and iterative development and low-cost prototyping techniques are essential. This research offers practical recommendations for startups and advocates for further studies on the evolution of AI design in various contexts over time.

Keywords— Human-Centered Design (HCD); Artificial Intelligence (AI); Minimum Viable Products (MVPs); Digital Startups; UX Design.

I. INTRODUCTION

Artificial Intelligence (AI) in digital products has transformed how digital startups think, evolve, and deliver value. In the early stages of product development, the Minimum Viable Product (MVP) is central to idea validation, user interaction, and iterative improvement based on feedback [1]. With AI tools and platforms being more accessible, even financially-strapped startups from developing countries like Pakistan are now participating in the experimentation of AI-supported MVPs—products that possess necessary but utilitarian AI components to demonstrate functionality and scalability. However, the black-box nature and lack of transparency in AI systems pose an atypical challenge to designers who aim to remain true to the canons of Human-Centered Design (HCD) at this early stage of product innovation [2].

Human-centered design prioritizes usability, accessibility, user empowerment, and ethical alignment in technology design. As capabilities like recommendation engines, chatbots, or predictive models are being introduced into MVPs, designers must render not only such capabilities technologically feasible but also comprehensible, reliable, and accessible [3]. However, aligning user-centric values with technical limits of AI like explainability, data quality, or algorithmic uncertainty can prove to be extremely challenging, particularly in the startup ecosystem of Pakistan, which is typically weak in technical infrastructure, skilled resources, and design maturity .

Additionally, MVPs are usually created under limited budgets and timelines with limited sets of features that are created for quick validation. Introducing AI can exacerbate problems like poor user research, a lack of interdisciplinarity, and unclear user expectations in this fast-turn, resource-scarce context. Designers, being often the interlocutors between engineering teams and end-users, have to operate under these limitations while ensuring that AI systems don't undermine user trust or perpetuate undesirable biases. These are exacerbated in environments like Pakistan, where digital literacy might be low and AI literacy is in its infancy [4].

Despite the increased application of AI in product design, little is known about the process by which designers create AI-powered MVPs, particularly in nascent digital economies. Most literature on this topic is biased towards either HCD or AI engineering in established tech economies, and there is a gap in research on the lived practices, design processes, and ethical challenges designers in nascent startups in South Asia undergo.

The research seeks to fill the gap by examining tensions and coping mechanisms of UI/UX and product designers creating AI-facilitated MVP design for Pakistani startups. Through a guided questionnaire, the research examines how professionals deal with tensions between technical complexity and human use, balance ethical responsibilities, and compromise design practice with local user needs. By placing the practitioner voice at the forefront, the research adds to the

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ongoing debate about responsible AI and inclusive technology design, while providing actionable recommendations for startups, educators, and policymakers interested in facilitating human-centric AI innovation in resource-poor settings.

II. LITERATURE REVIEW

A. Human-Centered Design (HCD) in the AI Context

Human-Centered Design (HCD) focuses on designing products based on user needs, desires, and limitations. Based on fields like psychology, interaction design, and participatory design, HCD has emerged as a foundational framework in product development, especially in digital interfaces and applications [5]. Yet, in the context of AI systems, HCD faces severe challenges. In contrast with conventional rule-based systems, AI responses frequently rely on data-driven estimates and black-box algorithms, which can be hard for users to understand, predict, or have faith in. Researchers stress the importance of developing AI interfaces that are explainable, transparent, and controllable, the features central to preserving user trust and usability [6].

The AI-HCD gap is more apparent in early-product development, when functionality is emphasized over elegance or profundity. Designers have to reconcile model sophistication with simplicity in user interfaces while keeping AI features from perpetuating bias, alienating users, or hiding functionality. This is especially true in resource-constrained environments, where data access, tools, and AI expertise are limited, constraining the design process [7].

B. Minimum Viable Products and AI-Powered MVPs

The Minimum Viable Product (MVP) is a lean startup principle that denotes a product with the bare minimum set of features to win over early adopters and prove core hypotheses. The MVP pattern is popular in digital startups as a way of accelerating time-to-market and experimenting with ideas fast [8]. Minimum Viable Products are typically designed with a lightweight structure and deliberately avoid complex backend development in the early stages. Nonetheless, the heightened availability of AI APIs (e.g., TensorFlow, GPT-4, OpenAI, Google Cloud AI) has resulted in the development of AI-enabled MVPs, in which startups implement rudimentary AI building blocks to prove fundamental value [9].

Studies indicate that, although AI can boost MVP capability (e.g., personalization, automation), it also presents data management, complexity, and ethical risk challenges. In the case of early-stage startups, AI integration tends to take place without proper testing or ethical review, thus risking poor usability and loss of user trust. Such implies the designer's responsibility in ensuring MVPs are human-centered, comprehensible, and iterative [10].

C. Design Challenges in AI Integration

Several design-specific challenges have been identified in recent literature regarding AI systems: Users often struggle to understand how AI makes decisions. They may either distrust or blindly rely on AI outputs, both of which are problematic. AI systems rely on user interaction to improve, but poor UI/UX can suppress meaningful feedback. Designers must navigate ethical dilemmas such as data bias and fairness, often without institutional support.

D. Pakistan's Digital Startup Ecosystem and Design Landscape

Pakistan's startup ecosystem has grown significantly in recent years, particularly in fintech, health tech, and edtech sectors. Many of these startups are now experimenting with AI capabilities to stand out in competitive markets. However, the design ecosystem in Pakistan still lacks maturity in terms of integrating AI with HCD principles. Digital literacy among users varies greatly, and infrastructural limitations constrain experimentation with advanced design frameworks. Moreover, designer–developer collaboration gaps and limited access to user research tools affect how AI-powered MVPs are conceptualized and tested.

There is a clear need to study how local designers navigate these constraints, especially when responsible for shaping user interactions with intelligent systems. Quantitative studies in this context are limited, and most available research remains theoretical or anecdotal.

III. RESEARCH METHODOLOGY

A. Research Design

The research used a quantitative, descriptive design to analyze the difficulties designers encounter when applying Human-Centered Design (HCD) to AI-based Minimum Viable Products (MVPs) in startups of new markets. Goal was to find out what factors and difficulties practitioners find in the early stages of creating AI products using standard measurement. Asking participants to respond using a questionnaire is considered best practice in HCI and design research when the focus is on measuring individual subjective experiences and their challenges with design [14]. Therefore same practice was utilized for this study.

Because the study looked at AI design challenges in startups, the researchers used a sampling method to focus on important points. This allowed them to get responses from people in Pakistan who worked on MVPs, like UI/UX designers, product managers, developers, and AI engineers.

B. Participants and Sampling Strategy

A total of 108 participants, primarily aged between 20 and 30 years, provided data for the study. Most of these individuals were early-career professionals working at digital startups, specifically in roles such as UI/UX Designers, Product Managers, Web/App Developers, AI Engineers, and Software Developers. All respondents reported having less than five years of experience in their current roles, with many indicating they had less than two years of experience in design or coding. This trend aligns with the typical background of innovators in the startup ecosystems of developing countries, where young, self-taught individuals drive change.

The purposive sampling method ensured that the selected participants had relevant experience that aligned with the study's objectives. This approach was effective as it allowed the research team to find individuals who were engaged in both design and AI within Minimum Viable Products (MVPs), which was essential for this particular study [15].

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C. Data Collection Instrument and procedure

The questionnaire was developed based on key constructs identified by previous studies on MVPs and HCD[16][17]. The questionnaire consisted of three sections. First section comprised of demographic data, second section asked respondents to rank the challenges they face while designing AI powered MVPs and third section asked about best practices needed to be adopted by practitioners while designing AI powered MVPs. Participants were asked to rate their experiences for each practice and challenge on a scale ranging from "Not significant" to "Extremely significant."

The challenges section addressed issues such as user understanding of AI, the quality of the data used, the values involved, the integration of AI with other technologies, and the level of trust users have in the system—drawing from prior research [16]. The practices section included examples such as early user participation, low-fidelity modeling, various tools that empower users to control decisions, and ethics assessments—all aligned with the principles of HCD and responsible AI studies [17].

To establish validity, the survey instrument was reviewed by three senior UX researchers familiar with AI design. A pilot test with 12 respondents was conducted to ensure clarity and consistency of the Likert-scale items. Minor changes were added to make sure the terms used are appropriate for Pakistan. Reliability analysis indicated satisfactory internal consistency (Cronbach's alpha = 0.68 for 'design challenges' and 0.72 for 'design practices')."

The data was collected from digital startups of Pakistan. The survey was shared using Google Forms.

IV. Results

A. Participant Demographics

The study gathered responses from young individuals and early-career workers involved in Pakistan's digital startups. A significant majority, 91.7% (n = 99), of respondents were aged between 20 and 30 years, while only 8.3% (n = 9) were under 20. This pattern indicates that startups are predominantly founded by young people, aligning with global trends that show innovations in AI-based products are primarily driven by recent graduates and younger individuals.

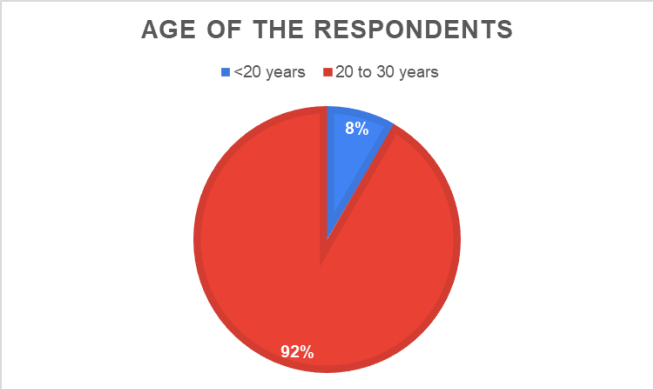


Fig. 1. Age distribution

The majority of respondents in the survey (98.1%) have had design experience for less than two years, while only a small percentage (1.9%) have been working in the field for 2 to 5 years. This indicates that many AI MVPs are being created by designers or developers who are still learning the fundamental principles of design. As a result, it can be more challenging for them to adhere to effective human-centered design (HCD) practices.

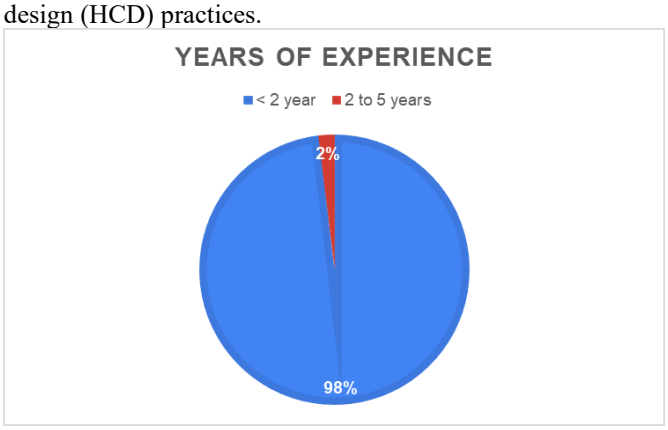


Fig. 2. Years of Design Experience

Regarding gender, more people in our survey were identified as male (54.6%) than female (45.4%), but still showing quite a balanced ratio for a technology field .

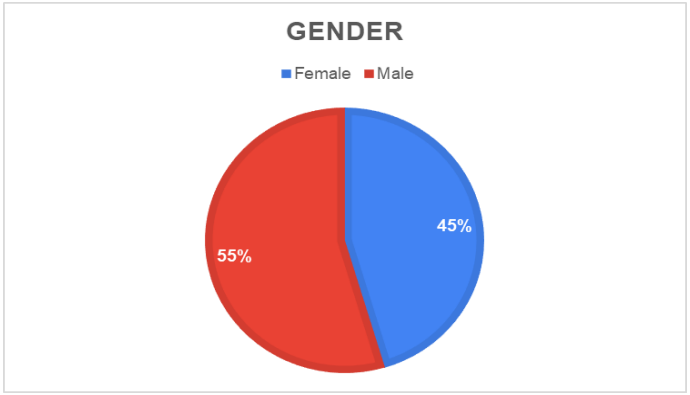


Fig. 3. Gender Distribution

B. Professional Backgrounds

UI/UX Designers represent the most common role, accounting for 37% (n = 40) of the sample. Following them, Product Managers make up 17.6% (n = 19) of respondents, while generalist designers or those with multiple roles comprise 14.8% (n = 16). The sample also included a significant number of software developers, app developers, backend developers, and AI developers. Additionally, about 8.3% of the total (n = 9) consisted of student designers, indicating an interest in AI design even before students enter the industry.

This diverse mix of roles provides a balanced perspective from designers, developers, and product managers, who typically influence decisions regarding the features included in a Minimum Viable Product (MVP) and its AI capabilities.

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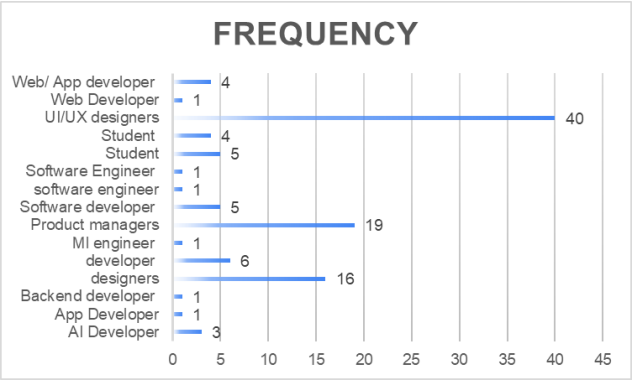


Fig. 4. Job Role Distribution

C. Challenges in AI-Integrated MVP Design

Participants were asked to evaluate the most significant challenges associated with using Human-Centered Design alongside AI when developing Minimum Viable Products (MVPs). They rated these challenges on a 5-point Likert scale, where 1 represented "Not significant" and 5 indicated "Extremely significant." The average scores highlight the primary issues affecting designers in today's landscape. The top 5 highest-rated challenges while designing AI powered MVPs are:

1. Integration issues between AI models and frontend design.
2. Limited availability of quality training data.
3. Low digital or AI literacy among target users
4. High cost of AI tools or platforms
5. Balancing simplicity with AI functionality

These challenges emphasizes that startups often struggle to obtain datasets that are clean, representative, and well-organized. This issue is particularly prevalent in economies with underdeveloped data ecosystems. If the training data for AI is inadequate, the models will not function accurately, negatively impacting user interaction with the product and introducing greater uncertainty in the design process. Secondly, the complexity of AI can conflict with the lean design principles of MVPs. Designers must find a balance between showcasing AI features and ensuring that users do not feel overwhelmed, especially when those users have little or no understanding of AI.

The statement "Explaining how AI features in MVPs work to users is a challenge" received a score of 2.8, indicating that users generally struggle to understand most AI functionalities in MVPs. If users do not trust AI-driven decisions, this can lead to ethical concerns, particularly when they lack comprehension of those decisions. The lowest-scoring challenge, at 2.6, was "It takes a lot of time to build working AI prototypes." While this remains a significant challenge, the score suggests that respondents view technical and data-related issues as more substantial than merely facing tight deadlines. This could imply that teams are already accustomed to producing rapid prototypes, and that technical challenges tend to slow them down more than the speed of coding.

D. Best Practices in AI-Integrated MVP Design

Participants were asked to prioritize the most significant practices associated with Human-Centered Design alongside AI when developing Minimum Viable Products (MVPs). They rated following practices as the most important ones on a 5-point Likert scale, where 1 represented "Not significant" and 5 indicated "Extremely significant." According to the ratings, participants identified these five practices to be most significant for successful design of AI powered MVPs:

1. Aligning AI functionalities with actual user needs
2. Conducting ethics reviews or discussions during design
3. Using low fidelity prototypes.
4. Iterative testing of AI features with real users.
5. Providing user control options over AI decisions

The results show that for a successful MVP design it is utmost important to understand user requirements and ensure there is no misuse of user data. It is equally important to work in iterations to incorporate feedback from users whether technical or non-technical to gain better understanding of user needs. To ensure minimal cost on MVP design , low fidelity prototypes can be used as an option

V. Discussion

The study aimed to understand the challenges and practices associated with applying Human-Centered Design (HCD) principles when developing AI-powered Minimum Viable Products (MVPs) in digital startups in Pakistan. Input from 108 participants revealed significant conflicts between what is technically feasible, what is ethical, and what users actually need during the early stages of product design.

A. Tensions Between AI Capability and Human-Centered Design

The research identified that the primary obstacles in AI-MVP development include integration issues between AI models and front end, a lack of high-quality data, low literacy of target users, high cost of AI tools and difficulties in balancing simplicity with AI features. These challenges underscore the issues that emerging markets face when building AI solutions.

For AI systems to function effectively, they require access to extensive amounts of clean, accurately labeled data. However, startups in Pakistan often lack well-defined data pipelines and data governance plans, resulting in outcomes that may be inconsistent or biased [18]. If AI systems are influenced by poor quality data, they may yield inaccurate or irrelevant results, frustrating users and diminishing their trust in the product [19]. This indicates a need for better access to ethical data collection tools, especially in resource-constrained environments.

Another significant difficulty in AI development is keeping the MVP simple. The concept of an MVP is to test a hypothesis with the least amount of work possible. However, AI functionality typically depends on deep user involvement, smooth interactions, and substantial IT efforts, which are rarely part of MVPs [20]. Therefore, AI should be presented in a user interface that is not overly complicated, particularly for users with limited digital knowledge [21]. This trade-off can lead to user confusion about why AI behaves in certain ways, ultimately reducing their likelihood of adopting the technology.

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B. Explainability and Trust as Emerging Themes

The complexity of many AI models makes it challenging for users to understand how the features work [22]. This issue is magnified in Pakistani startups, where end-users and sometimes design teams often lack sufficient knowledge about AI. Providing explanations of a model's workings—such as through visualized confidence or reasoning—can be beneficial, but only if users fully grasp the explanations.

Building user trust in AI systems is crucial in emerging markets, as many individuals remain wary of automation [21]. Trust encompasses users' feelings about their control, understanding, and empowerment in relation to the technology. The study found that many recommended practices from HCD frameworks for AI are seldom utilized, despite their potential benefits [16].

C. Resource Constraints and Design Marginalization

While time is recognized as a significant factor in startups, its role in building AI prototypes received a relatively low ranking of 2.6. Trends indicate that Pakistani startups often overlook design considerations during decision-making processes related to AI development [23]. Engineers and product leads tend to drive AI development from the outset, while designers are often brought in later, limiting their influence on fundamental decisions or ethic
al considerations.

The primary challenge lies not just in time constraints, but also in the lack of collaborative engagement between those developing AI and those designing user experiences. This observation aligns with findings from research conducted in the Global South [24]. Designers frequently have to navigate the complexities of algorithms and user needs without the necessary authority or funding to make impactful changes.

D. Implications for Design Practice in Emerging Markets

The findings of this research hold significant implications for enhancing AI utilization in resource-limited areas. First, it is essential to empower designers as well as the broader startup community. Design education in Pakistan should extend beyond HCD and usability to include foundational knowledge about AI, ethical considerations in data handling, and strategies for involving users in automated processes.

Additionally, encouraging the use of AI for low-fidelity prototyping—an area respondents indicated is rarely practiced—could promote experimentation with AI applications, allowing teams to test ideas cost-effectively.

Moreover, policy makers and funding agencies should ensure that AI frameworks prioritize ethics and human needs and are culturally appropriate. Approaches that may work for responsible AI in Western contexts do not always translate effectively to the challenges faced in South Asia.

E. Recommendations

Several recommendations aim to ensure that AI-powered minimum viable products (MVPs) are designed ethically, particularly for digital startups in Pakistan with limited resources.

First, it is essential for designers, AI developers, and product managers to collaborate from the very beginning of the product development process. By adopting collaborative approaches—such as holding joint design meetings and running sprints with diverse team members—communication issues can be resolved, ensuring that AI reflects human values.

Next, both design schools and companies should incorporate data literacy and AI literacy training into their curricula and employee development programs. Designers need a foundational understanding of how AI works, its limitations, and how to evaluate data for ethical considerations.

Startups should initially test AI features with simple prototypes and iterate on them based on feedback. This method allows design teams to observe user interactions with AI, gather valuable insights, and enhance both usability and trust without requiring extensive development resources. User onboarding for AI should be prioritized, especially when launching products in environments where users may not be familiar with AI technologies. AI designers can utilize tooltips, step-by-step guides, and clear prompts to explain AI functionalities and guide users in their actions.

To mitigate reliance on costly, proprietary AI solutions, funding organizations and policymakers should support the development of open-source datasets, ethical AI toolkits, and design standards that can be implemented locally.

F. Future Research

Future work should investigate how decisions made during the initial MVP phase can affect overall product development and user retention. Additionally, studies should explore regional differences in South Asia to identify various factors—such as cultural, regulatory, and infrastructural—that influence AI design. Employing techniques like interviews and observing how designers operate could also uncover ethical issues or challenges they face due to limited resources for AI innovation.

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